

## 5.6 Astrophysics and cosmology

This topic covers the physical interpretation of astronomical observations, the formation and evolution of stars, and the history and future of the universe.

Concept approach

| Students will be assessed on their ability to:                                                                                                                               | Suggested experiments |
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| 126 use the expression $F = Gm_1m_2/r^2$                                                                                                                                     |                       |
| 127 derive and use the expression $g = -Gm/r^2$ for the gravitational field due to a point mass                                                                              |                       |
| 128 recall similarities and differences between electric and gravitational fields                                                                                            |                       |
| 129 recognise and use the expression relating flux, luminosity and distance $F = L/4\pi d^2$<br><br>application to standard candles                                          |                       |
| 130 explain how distances can be determined using trigonometric parallax and by measurements on radiation flux received from objects of known luminosity (standard candles)  |                       |
| 131 recognise and use a simple Hertzsprung-Russell diagram to relate luminosity and temperature. Use this diagram to explain the life cycle of stars                         |                       |
| 132 recognise and use the expression $L = \sigma T^4 \times \text{surface area}$ , (for a sphere $L = 4\pi r^2 \sigma T^4$ ) (Stefan-Boltzmann law) for black body radiators |                       |
| 133 recognise and use the expression:<br>$\lambda_{\text{max}} T = 2.898 \times 10^{-3} \text{ m K}$ (Wien's law) for black body radiators                                   |                       |

| Students will be assessed on their ability to:                                                                                                                                                                                     | Suggested experiments |
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| 134 recognise and use the expressions<br>$z = \Delta\lambda/\lambda \approx \Delta f/f \approx v/c$ for a source of electromagnetic radiation moving relative to an observer and $v = H_0 d$ for objects at cosmological distances |                       |
| 135 be aware of the controversy over the age and ultimate fate of the universe associated with the value of the Hubble Constant and the possible existence of dark matter                                                          |                       |
| 136 explain the concept of nuclear binding energy, and recognise and use the expression $\Delta E = c^2 \Delta m$ and use the non SI atomic mass unit (u) in calculations of nuclear mass (including mass deficit) and energy      |                       |
| 137 describe the processes of nuclear fusion and fission                                                                                                                                                                           |                       |
| 138 explain the mechanism of nuclear fusion and the need for high densities of matter and high temperatures to bring it about and maintain it                                                                                      |                       |